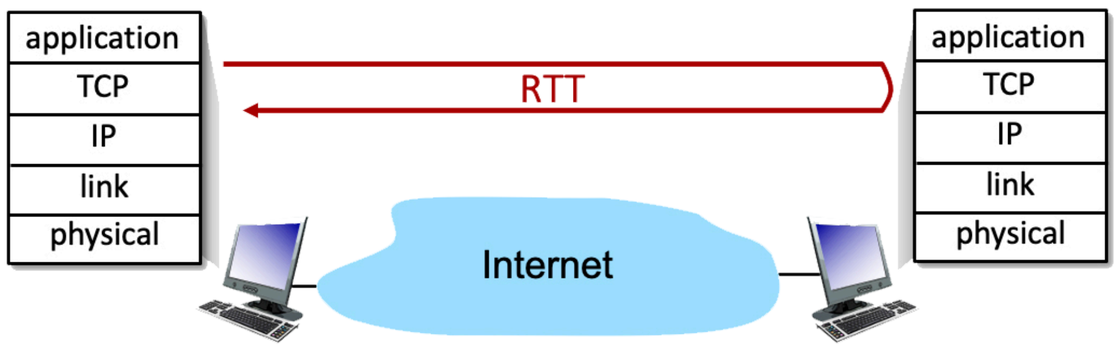
INTERACTIVE END-OF-CHAPTER EXERCISES

COMPUTING TCP'S RTT AND TIMEOUT VALUES

Suppose that TCP's current estimated values for the round trip time (*estimatedRTT*) and deviation in the RTT (*DevRTT*) are 210 msec and 39 msec, respectively (see Section 3.5.3 for a discussion of these variables). Suppose that the next three measured values of the RTT are 350 msec, 210 msec, and 370 msec respectively.



Compute TCP's new value of *DevRTT, estimatedRTT,* and the TCP timeout value after each of these three measured RTT values is obtained. Use the values of α = 0.125, and β = 0.25. Round your answers to two decimal places after leading zeros.

QUESTION LIST

1. What is the estimatedRTT after the first RTT?

2. What is the RTT Deviation for the the first RTT?

3. What is the TCP timeout for the first RTT?

4. What is the estimatedRTT after the second RTT?

5. What is the RTT Deviation for the the second RTT?

6. What is the TCP timeout for the second RTT?

7. What is the estimatedRTT after the third RTT?

8. What is the RTT Deviation for the the third RTT?

9. What is the TCP timeout for the third RTT?

SOLUTION

DevRTT is calculated with the following equation: (1-beta)*DevRTT + beta * |estimatedRTT - sampleRTT|
estimatedRTT is calculated with the following equation: (1-alpha)*estimatedRTT + alpha*sampleRTT
TCP timeout is calculated with the following equation: estimatedRTT + (4*DevRTT)

1. The estimatedRTT for RTT1 is 227.5

2. The DevRTT for RTT1 is 64.25

3. The timeout for RTT1 is 484.5

4. The estimatedRTT for RTT2 is 225.31

5. The DevRTT for RTT2 is 52.56

6. The timeout for RTT2 is 435.56

7. The estimatedRTT for RTT3 is 243.4

8. The DevRTT for RTT3 is 75.59

9. The timeout for RTT3 is 545.77

QUESTION 1 OF 9

What is the estimatedRTT after the first RTT?

Answer

CHECK | HINT | SKIP

We greatly appreciate the work of John Broderick (UMass '21) in helping to develop these interactive problems.

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Comments welcome and appreciated: kurose@cs.umass.edu